

PIPELINE WORKED EXAMPLE • STUDY CP-101

One deliverable, end to end — the Phase-1 PK package

Follow a single PK deliverable from Protocol to CSR — and see exactly what each stage routes to AI, to a validated tool, and to a human. The rule never changes: sensitivity decides the engine, validated tools own every reported number, and a statistician signs.

LOCAL

sensitive / unmasked

CLOUD (Claude, BAA)

hard, de-identified reasoning

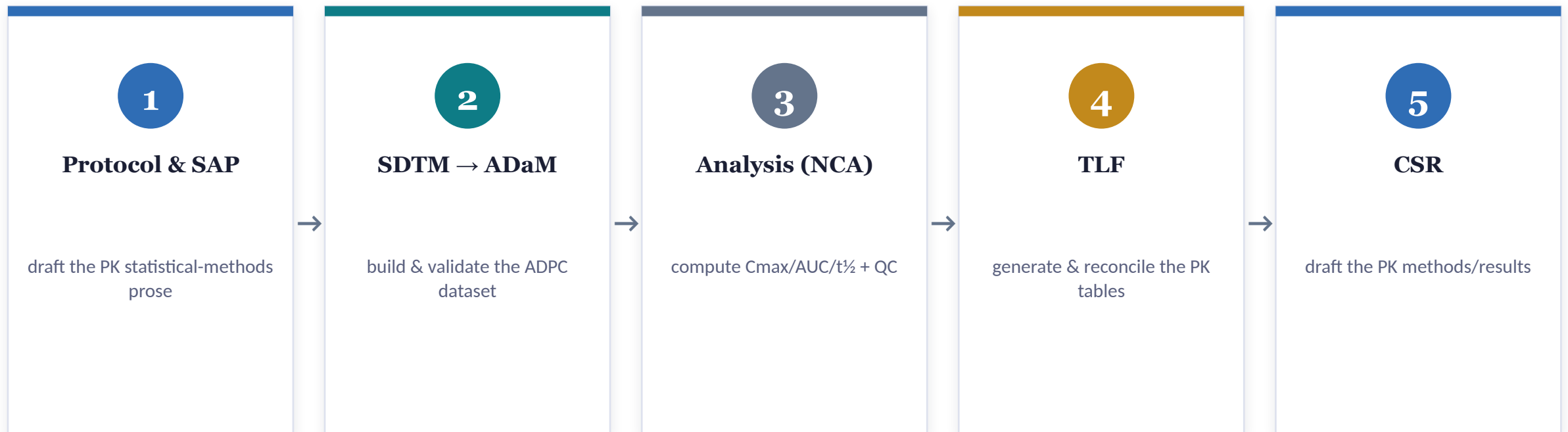
VALIDATED + HUMAN

the numbers & sign-off

Illustrative. Reported NCA numbers come from Phoenix WinNonlin, never an LLM.

The PK package = ADPC dataset + PK TLFs + the CSR PK section

Routing rule (every stage): touches unmasked / participant-level data → **LOCAL**; hard reasoning on de-identified text → **CLOUD Claude (BAA)**; every reported number → **validated tool**, QC'd and **signed by a human**.



Same deliverable, five stages — the engine changes with the data, but the numbers and the sign-off never leave validated tools and the statistician.

Drafting the PK statistical-methods section

AI ENGINE

CLOUD

Claude drafts the PK methods prose

From the protocol + SAP shell (de-identified text):
NCA parameters, BLQ rule, nominal-vs-actual time,
populations — with [placeholders] for study
values. Invents nothing.

VALIDATED TOOL — OWNS THE NUMBER

—

No reported number here yet; this stage is text.
The analysis logic stays authoritative in the SAP,
human-owned.

HUMAN — SIGNS

Statistician reviews & edits

Checks every derivation rule, fills placeholders,
owns the estimand and endpoint decisions. The
SAP is signed.

Hard, non-sensitive reasoning on de-identified text → cloud. The statistician owns the logic and the sign-off.

Building & validating the ADPC dataset (real PK data)

AI ENGINE

LOCAL

Local model assists the ADPC build

On the real, unmasked PC/PK data — zero egress: BLQ handling (pre-first-quantifiable→0, else LLOQ/2), nominal & actual time, deviation flags, mapping to ADPC. Drafts the spec & code.

VALIDATED TOOL — OWNS THE NUMBER

Pinnacle 21 validates conformance

CDISC/ADaMIG + Define-XML checks run in the validated tool — the conformance result is the tool's, not the model's.

HUMAN — SIGNS

QC double-programming; statistician signs

An independent program reconciles ADPC; discrepancies resolved; the dataset is signed and frozen.

Computing the PK parameters — the numbers

AI ENGINE

LOCAL

Local model assists QC & narrative

Sanity-checks profiles, flags outliers/anomalies for review, drafts the analysis notes — on the unmasked data, zero egress. It never produces a reported value.

VALIDATED TOOL — OWNS THE NUMBER

Phoenix WinNonlin computes the NCA

C_{max}, T_{max}, AUC_{0-t}, AUC_{0-inf}, t_½, CL/F, V_z/F — in the validated NCA tool. These ARE the reported numbers.

HUMAN — SIGNS

Independent double-programming; sign-off

A second analyst's run reconciles the parameters; the medical monitor reviews emerging PK; the statistician signs.

The reported NCA numbers come from Phoenix — never an LLM. The model assists QC; double-programming and a human own the result.

Generating & reconciling the PK tables, listings, figures

AI ENGINE

CLOUD

Model drafts shells, titles, footnotes

TLF shell prose, table titles & footnotes, and a consistency check of wording across outputs — text only.

VALIDATED TOOL — OWNS THE NUMBER

SAS / validated macros produce the TLFs

The actual tables/figures are produced by the validated programs from the frozen ADPC + NCA results.

HUMAN — SIGNS

QC reconciles to source; statistician signs

Independent QC ties every cell back to the validated outputs; the TLF package is signed.

The model drafts wording; the validated programs produce the numbers; QC reconciles to source and a human signs.

Drafting the PK methods & results section

AI ENGINE

CLOUD

Claude drafts the PK CSR prose

From the SAP + the (de-identified) summary outputs: a methods/results narrative consistent with the TLFs — cites the tables, paraphrases nothing numeric it shouldn't.

VALIDATED TOOL — OWNS THE NUMBER

The TLFs/Define remain the source of truth

Every number in the prose traces to a signed TLF; the model copies, never computes.

HUMAN — SIGNS

Medical writer + statistician finalize & sign

Human review for accuracy and interpretation; the CSR section is finalized into the regulated document.

De-identified summary reasoning → cloud draft. The signed TLFs own every number; humans finalize the regulated text.

AI did the drafting and the QC. Validated tools and humans owned the numbers.

Sensitivity routed the engine

Unmasked PK data stayed on the LOCAL model (zero egress); de-identified reasoning went to CLOUD Claude.

Validated tools owned every number

Phoenix computed the NCA; Pinnacle 21 validated conformance; SAS produced the TLFs. The LLM never produced a reported value.

A human signed every gate

Frozen ADPC, double-programmed parameters, reconciled TLFs, finalized CSR — each signed.

The payoff

Faster drafting and tighter QC across Protocol→CSR — with the data staying put and the regulated outputs unchanged in provenance.